

Pupillary Light Reflex Pathway:

- The light stimulates the retinal ganglionic cells. The impulses travel through the optic nerve (CN II), which projects bilaterally to the **pretectal nucleus** in the midbrain and then projects to the **Edinger-Westphal nucleus** bilaterally.
- The efferent limb is directed by the **oculomotor nerve (CN III)**. The Edinger-Westphal nucleus (preganglionic parasympathetic) relays to the **ciliary ganglion**.
- Miosis on the same side (direct light reflex) & opposite side (Consensual light reflex) to protect itself when light intensity increases.

Accommodation (near) reflex:

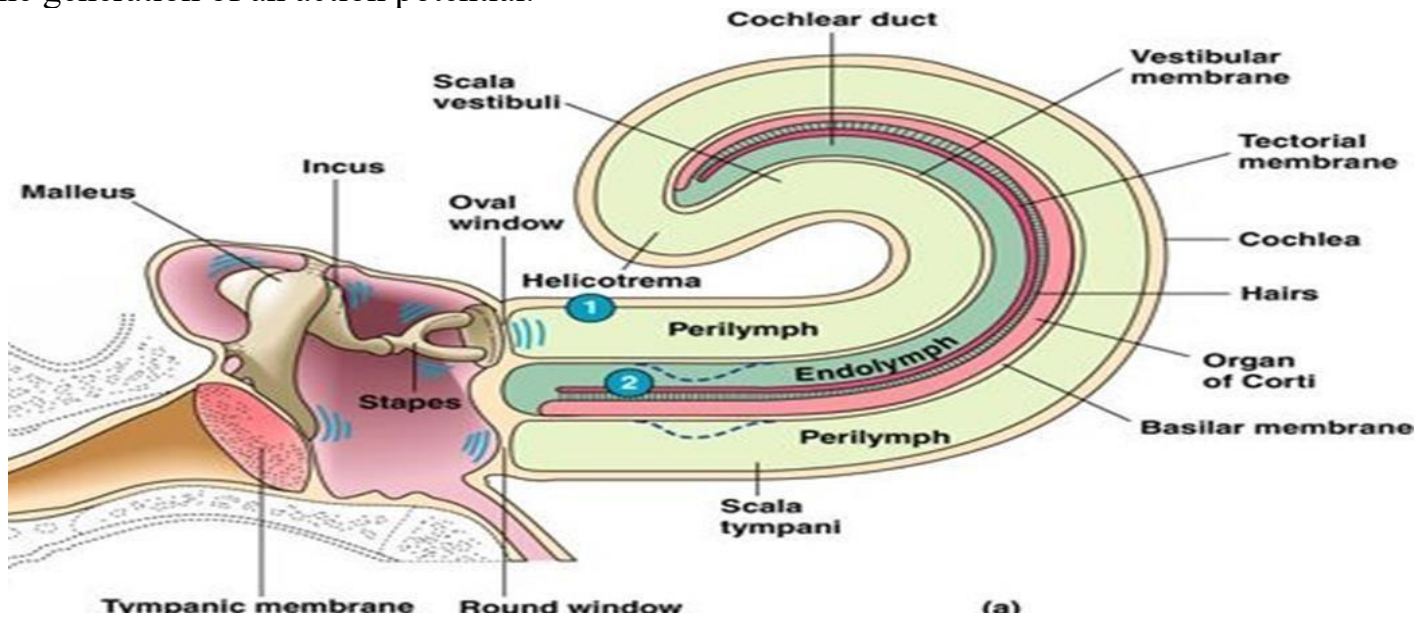
- The information is sent back through **efferent fibers** to the **pretectal nucleus** and then to the **Edinger-Westphal nucleus**, which contains preganglionic parasympathetic neurons whose axons travel in the oculomotor nerve to the orbit where they form a synapse in the **ciliary ganglion**.
- **Contraction of the sphincter pupillae and the ciliary muscle**
- **Convergent eye movements** caused by **contraction of the medial recti muscles** which are innervated by the **oculomotor nerve**.

Lecture (19): Auditory and vestibular pathways

Auditory pathway:

- ▣ The **ear pinna** and the **external auditory meatus** collect sound waves into a unidirectional wave towards the **tympanic membrane**.
- ▣ Sound waves cause vibrations of the **tympanic membrane**. The louder the sound the bigger the vibration, the lower pitch the sound the slower the vibration.
 - ▣ **Amplification** of sound waves occur in middle ear by **3 bony ossicles (malleus, incus & stapes)** which connect the **tympanic membrane** to the **oval window** towards the inner ear.
 - ▣ The **handle of the malleus** attaches with the tympanic membrane, and **the malleus** articulates with **the Incus**.
 - ▣ The **incus** articulates with the stapes, which impacts onto the **oval window**. The stapes moves like a piston, and causes the oval window to move in and out with sound waves.
 - ▣ In the inner ear, the sound waves are sent up the **scala vestibuli** to the apex of the cochlear duct (the helicotrema). Here it continues back down in the **scala tympani**.
 - ▣ The scala vestibuli and scala media are separated by **Reissner's membrane**.
 - ▣ Scala media and scala tympani below are separated by the **basilar membrane**.

- When sound waves move up and down the **perilymph** in the **scala vestibuli** and **scala tympani**, the **vibrations** move the **basilar membrane**.
- The **organ of Corti (receptor of hearing)** lies on the **basilar membrane**, and is responsible for converting vibrations into **neural signals**.
- When vibrations move the basilar membrane, the **stereocilia** that lie on the **hair cells** of organ of Corti bend on the **tectorial membrane**, and potassium channels open. The influx of potassium causes the generation of an action potential.



- Signals from organ of Corti are sent up the **spiral ganglion** in the cochlea.
- The **Spiral ganglion** houses the cell bodies of the **first order neurons**. Their central axons form the **cochlear nerve**.
- The **cochlear nerve** enters the **cranium** through the **internal acoustic meatus** and travels a short distance to enter the brainstem at the **cerebellopontine angle**.
- Then it sends the signals to the **ventral and dorsal cochlear nuclei** in the brainstem.

These nuclei are the first central connection with the auditory information.

- Most fibres from the **dorsal cochlear nucleus**, cross the midline and ascend in the contralateral lateral lemniscus. Other fibres ascend in the ipsilateral lateral lemniscus.

- Most fibres from the **ventral cochlear nucleus** **decussate** to the **contralateral superior olivary nuclei** in a region known as the **trapezoid body**. Some fibres synapse at the ipsilateral superior olivary nucleus.

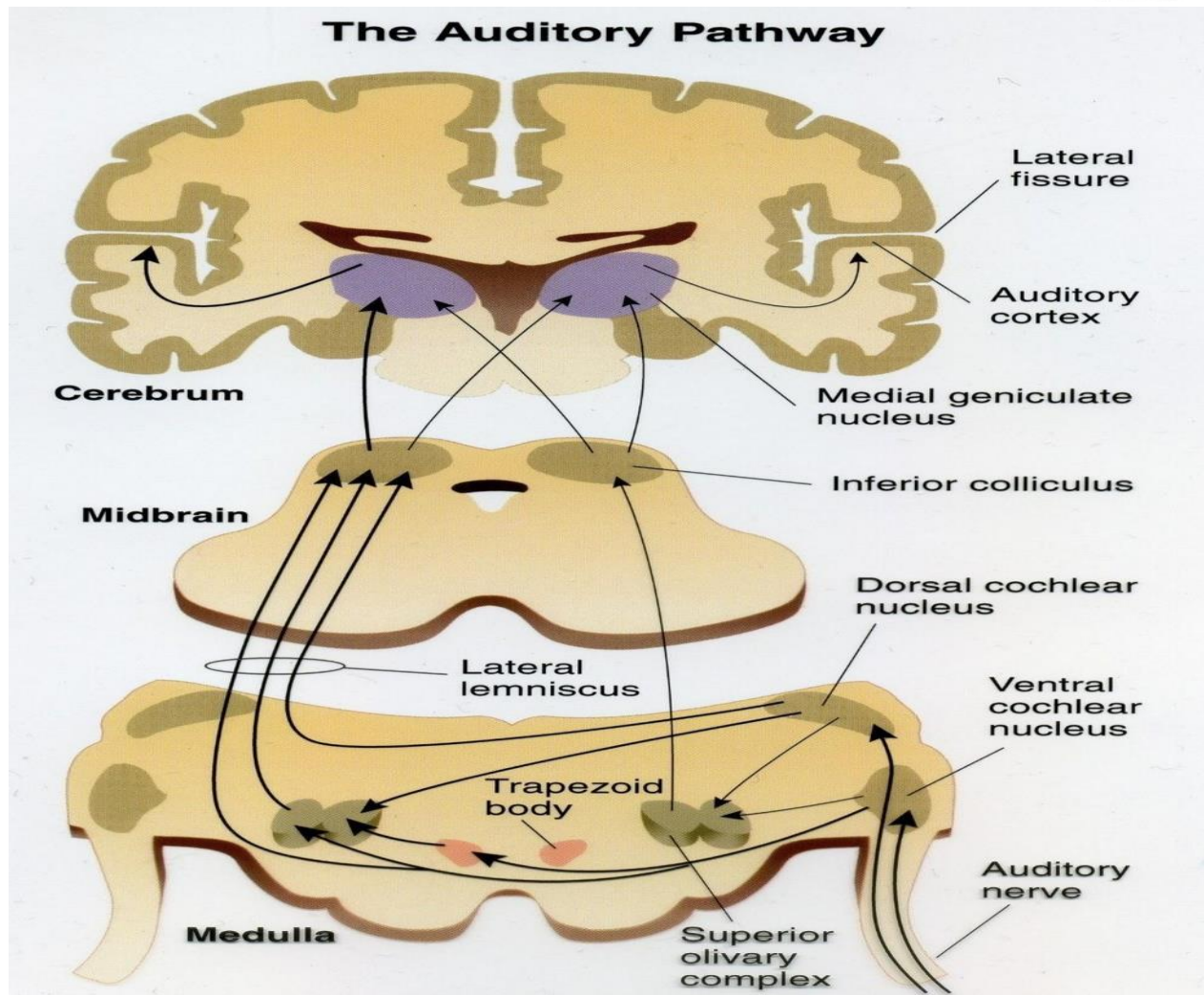
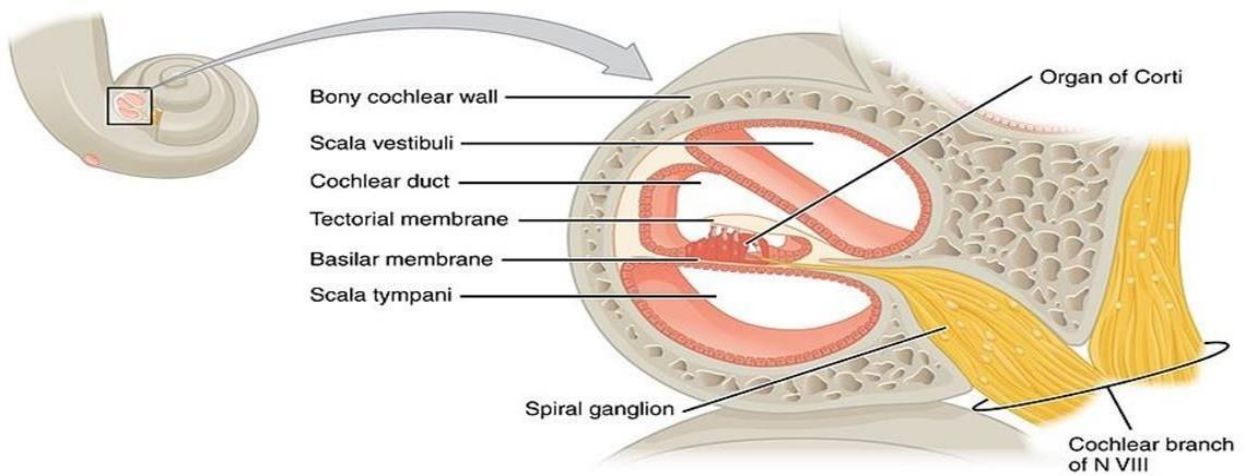
- The **outputs** of these nuclei are to:

- 1- The **superior olivary nucleus** (via the **trapezoid body**). The other half of the information is sent to the contralateral superior olivary complex.

2- The lateral lemniscus then to the inferior colliculus.

3- The majority of these connections will ultimately terminate in the **auditory cortex**.

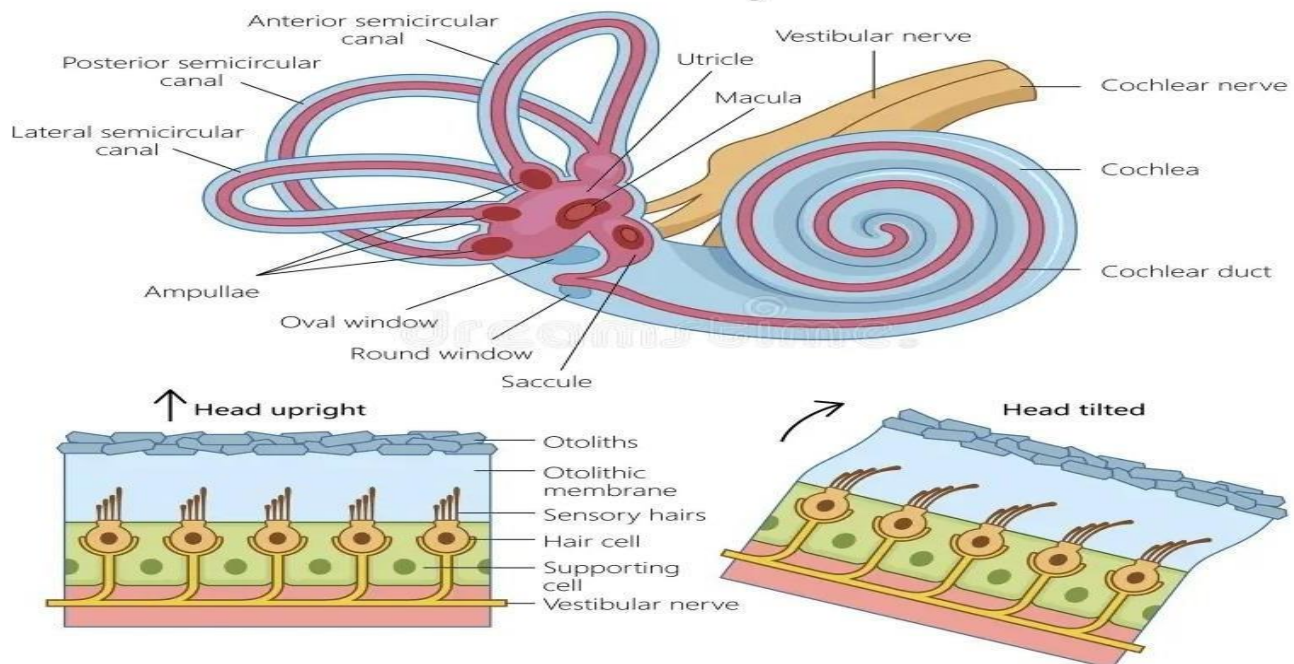
- The **superior olivary nucleus** is located just next to the trapezoid body in the pons. It also projects upwards through the lateral lemniscus. It has a number of roles in the process of hearing.
- **First station for binaural hearing.**
- **Localization and direction of sounds.**
- The **lateral lemniscus**:
 - It is the ascending brainstem auditory pathway.
 - Carries axons from the contralateral dorsal and ventral cochlear nuclei and axons of superior olivary nucleus to the inferior colliculus.
- The **inferior colliculus**:
 - It is located on the dorsal aspect of the midbrain.
 - It projects to the medial geniculate nucleus of the thalamus via inferior brachium of the midbrain.
 - Vertical and horizontal sound source localization.
- The **medial geniculate nucleus**:
 - It is located on the inferior surface of thalamus
 - This is the thalamic auditory relay nucleus between the inferior colliculus and the auditory cortex.
 - Its axons form **auditory radiation** which pass through the **sublenticular** part of the **internal capsule** to end on the middle of superior temporal gyrus.
- The **primary auditory cortex (Area 41, 42)**:
 - is in the **middle of superior temporal gyrus of the temporal lobe**.
 - Conscious perception of sounds and processing of auditory information.
- **The secondary auditory cortex (areas 22, 39, 40)**: This is a region on the temporal-parietal junction and on the left side of the brain, which is responsible for **understanding speech**.

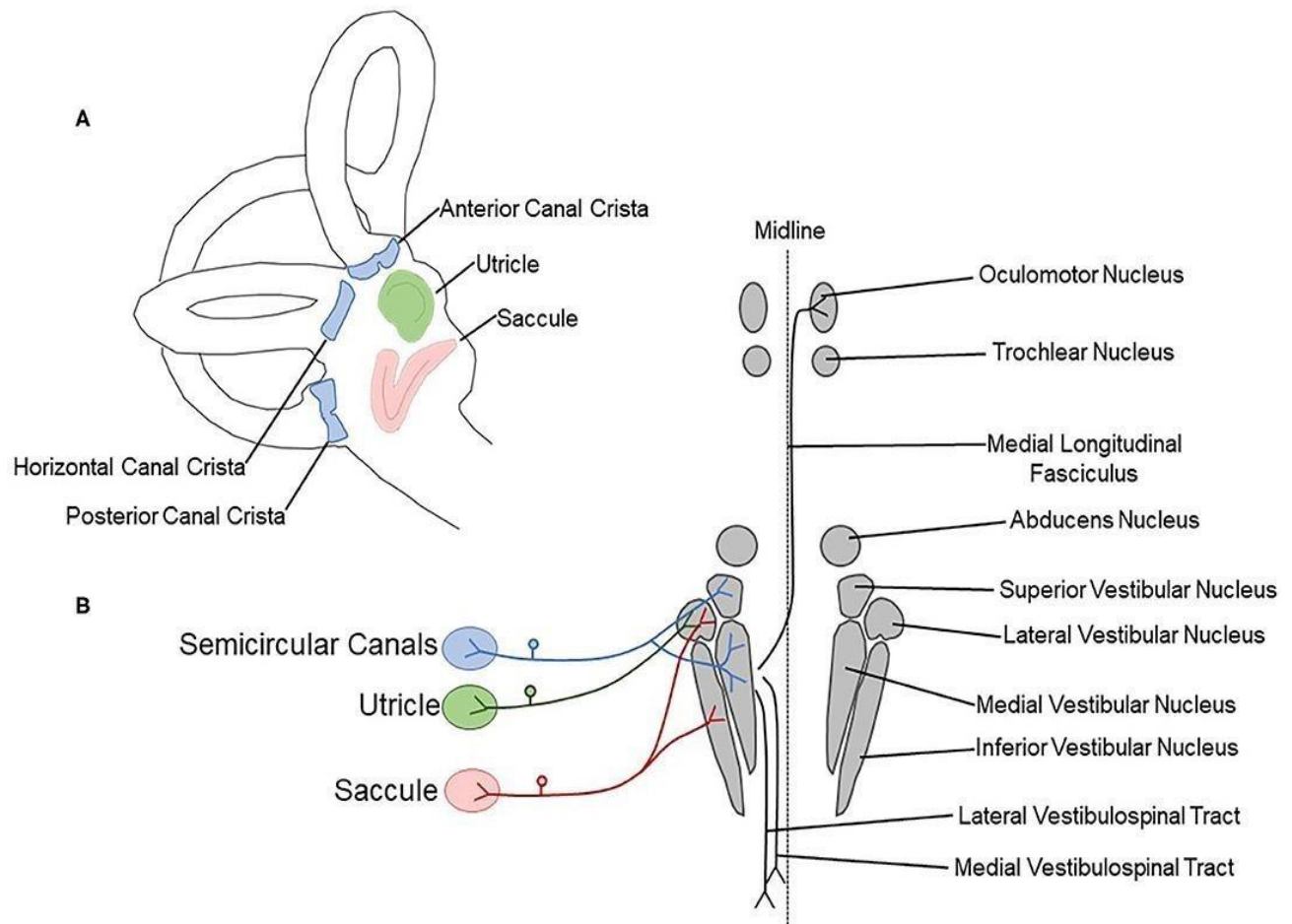


Vestibular Pathway

- **Receptors:** are the hair cells located in the membranous labyrinth.
- **Macula utricle** and **macula saccule** in **vestibule**, these receptors are responded to linear acceleration and deceleration of the head.
- **3 cristae ampullares in 3 semicircular canals** which respond to **angular or rotatory** movements of the head.
- **1st order neuron: vestibular ganglion (bipolar neuron)**, its peripheral processes end in receptors, its central processes join the vestibular nerve→ end in:
 - **2nd order neuron:** in the 4 vestibular nuclei (lateral, medial, inferior and lateral).
 - Some fibers go directly to the **cerebellum** through the inferior cerebellar peduncle.
- **Efferents from the vestibular nuclei project to:**
 - **Flocculonodular lobe of cerebellum** through inferior cerebellar peduncle.
 - **Motor nuclei of cranial nerves** through medial longitudinal fasciculus.
 - **Spinal cord as lateral & medial vestibulospinal tracts.**
- **3rd order neuron: Ventral posterior medial nucleus of the thalamus (PMVNT)**, which in turn projects to the cerebral cortex.
- **Ocular cranial nerve nuclei (3,4,6)** for conjugate movements of the eye.
- **Vestibular Cortex:** The cortical region responsible for conscious awareness of vestibular sensation, adjacent to head area of the sensory cortex in parietal lobe.

Vestibular system





• Central Vestibular Pathways

